PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for

B.Voc. (Medical Laboratory and Molecular Diagnostic Technology)

as per NEP 2020

W.e.f. session2024-25

Course Outline for B.Voc (MLMDT)
as per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLMDT 101	Core Discipline	Basics of Human Anatomy	2	2	4	40	10	40	10	100
MLMDT 102	Core Discipline	Introduction to Laboratory Equipments	2	2	4	40	10	40	10	100
MLMDT 103	Core Discipline	Introduction to Hematology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLMDT 201	Core Discipline	General Pathology	2	2	4	40	10	40	10	100
MLMDT 202	Core Discipline	Basics of Biochemistry	2	2	4	40	10	40	10	100
MLMDT 203	Core Discipline	Analytical Laboratory Testing Process	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Medical Lab Technologist

*SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter2ndSemesterExaminations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3weeks of the start of 3rdsemester.

** Minimum Credits required to get Certificate Course in B. Voc. (MLMDT) I = 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (MLMDT) (SEMESTERS I & II)

PROGRAMME OUTCOMES

PO1. Understanding the concept, principles of working of the Medical and Molecular Diagnostic lab**.**

PO2. Understanding the Human anatomy and various diseases related to human body**.**

PO3. Understanding the concept and procedures of various Analytical and Biochemical test**.**

PO4. To develop ability to work in the Hospital and understand the Healthcare system.

PO5. To develop awareness of the Marketing strategies, Management hierarchy involved in various labs

PO6. To develop the skill and application of Information Technology in the field of biosciences

Semester I

	COURSE TITLE	BASICS OF HUMAN ANATOMY
1.	**Course Code**	Skill MLMDT 101
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab Technologist.
7.	**Course Objectives**	Basic understanding of organization of body cells, tissues, organs, organ systems, and glands in human body.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1.Gain the sufficient knowledge of cellular structure and its working. CO2. Understanding of tissue system, organs and organ system. CO3.Learning the cellular communication through receptor studies.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.
11.	**Course Content**	**UNIT– I** Basic unit of body - Cell The anatomical organization of body cells, tissues, organs, organ systems, membranes and glands in human body Introduction to different types of tissues: Anatomy, epithelial tissue, muscular tissue, nervous tissue Different types of organ Brief Introduction of different types of body fluids, secretions and excretions Skeletal system: Bones, joints and muscles

UNIT– II

Digestive Organs:

- Teeth
- Tongue
- Oral cavity
- Pharynx
- Oesophagus
- Stomach
- Small intestine
- Large intestine
- Liver, Pancreas and Spleen

UNIT– III

Respiratory Organs:

- Nasopharynx
- Oropharynx
- Larynx
- Trachea
- Bronchi
- Lungs
- Thoracic cavity
- Pleura and Pleural cavity

Circulation System: Structure of Heart and Brief introduction of main blood vessels

UNIT– IV

Reproductive Organs**:** Male and Female Gonads

Nervous system and Sense organs: Brief Introduction of Central Nervous System and Peripheral Nervous System Anatomy of Brain, Spinal Cord, Nerves, Eye, Ear,
Olfactory Receptors, Gustatory Receptors

Excretory Organs:

- Cortex and medulla of Kidney
- Urinary Bladder
- Urethra: Male, female

Practical Work:

1. Preparation and study of mitosis in onion root tips
2. To demonstrate the presence of Barr body from the oral mucosa of human female
3. Study of various organ through charts and models.

12. Suggested Resources for Reading:

1. Ross and Wilson, Anatomy and Physiology in Health & Illness
2. Murgesh, N., Basic Anatomy and Physiology, Sathya Publishers
3. Pearce, E., Anatomy and Physiology for Nurses, Jaypee Brothers
4. O' Rahilly, R.R., Basic Human Anatomy: Regional Study of Human Structure, WB Saunders Co. Ltd.
5. Sears, W.G., Winwood, R.S. & Smile, J.L., Anatomy and Physiology for Nurses and Students of Human Biology, Hodder Arnold

Note: Latest edition of text books may be used.

	COURSE TITLE	INTRODUCTION TO LABORATORY EQUIPMENTS
1.	**Course Code**	MLMDT 102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab Technologist.
7.	**Course Objectives**	It provides opportunity to perform hands on practical training and Laboratory equipment.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. To gain broad understanding of care of laboratory glassware, equipment and instrument. CO2. To gain broad understanding of setting up, calibrating, operating, cleaning, maintaining, and troubleshooting of laboratory equipment used in quantitative or qualitative analysis. CO3. To Calibrate and Validate the Clinical Laboratory instruments and glass wares. CO4. To understand Microscopy, working principle, maintenance and applications of various types of microscopes
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.
11.	**Course Content**	**UNIT– I** Brief understanding of laboratory planning and laboratory key learning outcome operations Introduction to operation and safety precautions of common Laboratory Equipments Principles and working of laboratory instruments: Incubator, Hot Air Oven, Water Bath, Anaerobic Jar, Centrifuge, & pipettes, colorimeter, Neubauer's Chamber Setting up, calibrating and validating, operating, cleaning, maintaining

Autoclave – Its structure, functioning, control and indicator Micrometry
Elementary Knowledge of Analytical Chemistry: Balances- monopan, two pan, top pan, pH meter, spectrophotometer, fluorimeter, flame photometer, ion selective electrodes, urinometer

UNIT– II

Techniques of Disinfection
Sterilization: Definition, Classification and General Principle of Sterilization, Sterilization Techniques, Sterilization of rubber goods, laboratory equipment & other instruments.
The cleaning and maintenance procedures of the machine
Identify the cause of errors or other problems or defects in Equipments
Glassware – Description of Glassware, its use, handling, and cleaning of glass ware

UNIT– III

To study the compound microscope and its parts
Microscopy: Working principle, maintenance, and applications of various types of microscopes. Care and handling of various microscopes
Dark ground microscope
Polarizing microscope
Phase contrast microscope
Interference microscope
U.V. light microscope
Electron microscope

UNIT-IV

Ultra-microtomy
Importance of conservation of resources in laboratories
Museum techniques
Lab safety and Instrumentation

Practical Work:

1. Demonstration of the use of autoclave, centrifuge, and spectrophotometer
2. Cleaning and sterilization of glassware using hot air oven, autoclave etc.
3. Calibration of apparatus and glassware
4. Demonstration of various equipments for the estimation of hemoglobin, WBC, RBC (Neubauer chamber setting up)
5. Demonstration of parts of simple and compound microscope, phase contrast microscope, its functioning and care
6. Demonstration of safety rules in laboratory like proper handling of patients specimens and disposal of syringes, needles etc.

12. Suggested Resources for Reading

1. At the Helm: A Laboratory Navigator, Barker, K., Cold Spring Harber Laboratory Press Basic Medical Laboratory Techniques, Estridge, B.H., Walters, N.J. & Reynold, A. Cengage Learning Chatwal, G.R., & Anand, S., Instrumental Methods of Chemical Analysis, HpH
2. Roskans, J. & Rodgers, L., Lab Ref: A Handbook of Recipes, Reagent and Other Reference Tools for Use at the Bench, Cold Spring Harber Laboratory Press
3. Haminond, M., Oxford Concise Medical Dictionary, Oxford University Press

Note: Latest edition of text books may be used.

	COURSE TITLE	INTRODUCTION TO HEMATOLOGY
1.	**Course Code**	MLMDT 103
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab Technologist.
7.	**Course Objectives**	To gain understanding of blood and components of blood to gain knowledge of hematological Diseases and hematological investigations.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Learn the various component of blood. CO2. Understand the blood circulation, coagulation, and haemostatis. CO3.Understand the various haemodynamic disorder.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.
11.	**Course Content**	**UNIT– I** Introduction to hematology and laboratory organization. Composition and functions of blood and lymph Detailed study of Hemoglobin and functions of hemoglobin Blood groups including Rh. Factor Detailed study of Reticulocytes Formation of blood. Morphology of normal blood cells and their identifications **UNIT– II** Hemostasis– Definition, Basic concept and principle, Basic steps involved in Hemostasis. Coagulation– Basic Physiology, coagulation factors, Fibrinolysis Various anticoagulants, their uses, mode of action and their merits and demerits. Normal and absolute values in hematology Quality assurance in hematology

UNIT– III

Descriptive study of RBC abnormalities, Disorders related to RBC Normal white cell count & physiological variation, Normal white cell count & physiological variation. Hematological Diseases: Anemia and various types of anemia, Thalassemia, Polycythemia, hemolytic disease of new born, multiple myeloma.

Leukemia: Definition and classification

Introduction to Bleeding Disorders

UNIT-IV

Laboratory preparation for coagulation tests, Routine coagulation tests, prothrombin time, plasma recalcification time, partial thromboplastin time, activated partial thromboplastin time, thrombin time.

Practical Work:

1. Cleaning of Laboratory glassware in haematology.
2. Clinical significance, specimen collection, laboratory investigation & preservation of blood for various haematological investigations.
3. Preparation of blood smears
4. Total leukocyte counts and Differential leukocyte count
5. Determination of total erythrocyte (RBC) count and platelet counts. Errors involved and mean to minimize such errors
6. Haemoglobinometry, various methods of estimation of Hb, errors involved and standardization of instrument for adaptation for Hb estimation.
7. Romanowsky dyes, preparation and staining procedures of blood smears
8. Laboratory tests for assessing bleeding disorders
9. Determination of erythrocyte sedimentation rate (ESR)

12. Suggested Resources for Reading

1. Atlas of Hematology, McDonald, G.A., Cruickshank, B. & Paul, J. Churchill
2. Clinical Hematology, Ludlam, C.A., Churchill Living Stone, Living Stone
3. Bain, B.J., Bates, I., & Laffan, M.A., Dacie and Levis, Practical Hematology Elsevier
4. Practical Hematology (International edition), Dacie, J.V. & Lavie, S, M., Churchill Living Stone
5. Hematology, Besa, E.C., Lippincott, Williams & Wilkins
6. Hematology: Pathophysiological Basis for Clinical Diagnosis, Reich, P.R. & Robinson, S.H., Lippincott Williams & Wilkins
7. Hematology for Students and Practitioners, Sood, R., Jaypee Brothers Medical Publishers
8. Hand Book of Medical Laboratory Technology, V.H., CBJ

Note: Latest edition of text books may be used.

Semester II

	COURSE TITLE	GENERAL PATHOLOGY
1.	**Course Code**	MLMDT 201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab technologist.
7.	**Course Objectives**	Understanding the general principles, terminology, diagnostic procedure and basic concept of pathology.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Study cell injury, death and adaptation. CO2. Understand the concept of cancer and tumor. CO3. Get knowledge of genetic and paediatric diseases.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.
11.	**Course Content**	**UNIT– I** Normal Cell Cell Injury: Types of cell injury, Etiology of cell injury, morphology of cell injury, Cellular swelling Cell death: Types- Autolysis, necrosis, apoptosis & gangrene Cellular adaptations: Atrophy, hypertrophy, hyperplasia & dysplasia Acute inflammation, Vascular events, cellular events, Inflammatory cells, Morphology of acute inflammation, Chronic inflammation Regeneration, repairs, wound healing and Healing in specialized tissues

UNIT– II

Hemodynamic disorders: Internal environment, Normal water and electrolyte balance, Disturbances of body fluids and
electrolytes, Oedema
Disturbances in volume of circulating blood: Hyperemia and congestion, Hemorrhage and shock
Circulatory disturbances of obstructive nature: Thrombosis, Ischemia, Infarction

UNIT– III

Neoplasia: Nomenclature and classification, Characteristics of tumors, Local invasion and metastasis
Prognostic markers, Grading and staging of cancer
Viruses and tumor
Clinical aspects of neoplasia, Tumor host interrelationship

UNIT-IV

Diagnosis of cance
Genetic and pediatric diseases: Introduction to Genetic diseases, Developmental defects and Cytogenetic abnormalities
Nutritional diseases: Obesity, Starvation, Protein energy malnutrition, Disorders of vitamins and Trace element

Pathology:

1. To study cultural isolation and identify of pus miscrobes from wound.
2. Urine analysis
3. To identify pus microbes at genetic level
4. Study of various microscopes

12. Suggested Resources for Reading

1. Mohan, H., Textbook of Pathology, Jaypee Brothers Medical Publishers
2. Abbas, A.K., Andra, T.D., Astu, J.C. & Kumar, V., Basic Pathology
3. Cree, I.A., Pathology: A comprehensive Review, CRC Press
4. Strayer, D.S., & Saffitz, J.E., Rubin's Pathology: Mechanism of Human Disease, Lippincott, Williams & Williams

Note: Latest edition of text books may be used.

	COURSE TITLE	BASICS OF BIOCHEMISTRY
1.	**Course Code**	MLMDT 202
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP** (Lectures) (Practical)	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab Technologist.
7.	**Course Objectives**	To gain elementary knowledge of Biochemistry.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Get knowledge of inorganic chemistry and preparation of solution and buffers. CO2. Understand organic compound, their structure and properties. CO3.Detailed knowledge of various biomolecules like Carbohydrates, Lipids, Proteins and Nucleic acid.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.
11.	**Course Content**	**UNIT– I** Elementary Knowledge of Inorganic Chemistry: Role and properties of Water, Acids, bases and salts, Acid Base Balance, pH, Henderson- Hasselbalch equation. Normality, Molarity and Molality, Mole, Molar solutions, normal solutions, buffer solutions, physiological buffers, percent solutions, saturated solutions, standard solutions

UNIT– II

Elementary Knowledge of Organic Chemistry: Organic compounds- Alcohol, Aldehyde, Ketone and Carboxylic acid: General Structure, Nomenclature, Classification and Physical Properties.

Elementary Knowledge of Physical Chemistry- Osmosis, osmotic pressure, dialysis, surface tension, diffusion, hypotonic, hypertonic, and isotonic solutions. Definition and classification of some colloids and crystalloids

UNIT– III

Carbohydrates: Definition, Classification, Biomedical importance & properties.

Amino Acids and Proteins: Definition, Classification and Biomedical importance & properties.

UNIT-IV

Lipids: Definition, Classification and Biomedical importance, Fatty liver, Cholesterol & its clinical significance, Lipoproteins in the blood & their functions.

Introduction and general properties of Nucleic Acids Definition, Basic structure of DNA and RNA, types of RNA, Functions of nucleic acids.

Practical Work:

1. Preparation of buffer solutions and measurement of their pH
2. Verification of Beer Lambert's Law
3. To prepare the different concentrations of solutions
4. To find out the normality of the given solution
5. Qualitative tests for:
   (a) Carbohydrates.
   (b) Amino acids and proteins,
6. Estimation of ascorbic acid by dye method.
7. Estimation of Amino acids by ninhydrin method.
8. Estimation of Protein by biuret & Lowry method.

12. Suggested Resources for Reading

1. A Manual of Laboratory & Diagnostic Tests, Frances Fischbach
2. Biochemistry, Voet and Voet
3. Biochemistry, Stryer
4. Biochemistry, U. Satyanarayan. & U. Chakrapan
5. Clinical Biochemistry, Richard Luxton
6. Clinical Diagnosis & Management by Laboratory Methods, John Bernard Henary
7. Clinical Biochemistry, G. Guru
8. Handbook of Biochemistry, M.A. Siddique
9. Textbook of Medical Biochemistry, S. Ramkrishnan
10. Biochemical Techniques, K. Choudhary
11. Text book of Medical Biochemistry, Chaterjee & Shinde
12. Principles of Biochemistry, David L. Nelson
13. Principles of Biochemistry, Lehninger
14. Textbook of Biochemistry and Human Biology, G.P. Talwar
15. Textbook of Medical Laboratory Technology, Godkar and Godkar
16. Outline of Biochemistry, Conn Stumpf
17. Principles of Internal Medicine, Issel bacher
18. Proteins and Proteomics: Laboratory Manual, Richard J. Simpson
19. Purifying Proteins for Proteomics: Laboratory Manual, Richard J. Simpson
20. Enzymes: Biochemistry, Biotechnology & Clinical chemistry, Palmer Trevor, Publisher: Horwood Pub. Co.

21. Outlines of Biochemistry, Erice Conn & Paul Stumpf; John Wiley and Sons
22. Fundamentals of Biochemistry, Donald Voet & Judith Voet, John Wiley and Sons, Inc
23. Lehninger, Principles of Biochemistry, David Nelson & Michael Cox, W.H. Freeman and company
24. Biochemistry, Jeremy Berg, Lubert Stryer, W.H. Freeman and company
25. Biochemical Methods for Agricultural Sciences – Sadasivam and Manikam. Wiley Eastern Limited
26. Practical Clinical Biochemistry Harold Varley, CBS
27. An Introduction to Practical Biochemistry, David T Plummer. Tata McGraw-Hill Publishing Company Limited,
28. Biometrics Identity, Sameer Nanawati
29. Medicinal Chemistry, Ashutosh Kar

Note: Latest edition of text books may be used.

	COURSE TITLE	ANALYTICAL LABORATORY TESTING PROCESS
1.	**Course Code**	MLMDT 203
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Medical Lab Technology.
7.	**Course Objectives**	To gain broad understanding of chemicals/reagents and procedure useful in sample analysis.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1.learn about collection and maintaining records of biological specimen and various body fluid sample. CO2. gain broad knowledge of performing various routine haematological tests, urine tests, stool tests, semen tests and sputum tests. CO3. Interpret the results of the above test for further diagnosis.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.
11.	**Course Content**	**UNIT– I** Collection and recording of biological specimens. Collection, Handling and transportation of Blood Sample Routine Hematological Tests Examinations of sample for serum, Urea and Uric acid. Estimation of Essential electrolytes: Sodium, potassium, calcium, chloride and phosphorus Determination of blood sugar level of plasma (or serum): Ortho-toluidine method and Glucose oxidase method. Determination of serum urea nitrogen: Diacetyl monoxime method. Determination of serum creatinine: Alkaline picrate method. Determination of serum total cholesterol

UNIT– II

Determination of serum bilirubin: Malloy and Evelyn method and DMSO method
Laboratory investigation for anaemia
Laboratory investigation for disseminated intravascular coagulation (DIC)
Mechanism of fibrinolysis test for fibrinolysis. Platelet function tests and their interpretation
Erythrocyte sedimentation rate (ESR), factors influencing ESR and various procedures for its estimation with their significance. Determination of pack cell volume (PCV). Physiological variations in Hb, PCV, TLC and platelets
Determination and calculation of red blood indices: MCV, MCH, MCHC

UNIT– III

Determination of absolute Eosinophil count and Reticulocyte count
Automated systems in hematology
Semen analysis: Composition of semen, Collection and preservation of semen, Physical and chemical examination of semen, microscopic examination of semen, Clinical significance of semen analysis
Sputum analysis: Composition of sputum, Collection and preservation of sputum, Physical and chemical examination of sputum, microscopic examination of sputum, Clinical significance of sputum analysis

UNIT-IV

Urine Examination**:**

- Collection, Handling and transportation of Urine
- Urine analysis, routine examination of urine (physical examination of urine), rapid chemical tests of Urine
- Determination of specific gravity of urine by urinometer and refractometer
- Biochemical Test Profile (Quantitative determination of Urine): Amylase, Calcium, Chlorides, Creatinine, Sodium, Potassium, Glucose, Proteins, Urea nitrogen, uric acid

Practical Work:

1. All Biochemical tests mentioned in Theory

12. Suggested Resources for Reading

1. Textbook of Medical Laboratory Technology, Godkar and Godkar
2. Research Methodology in Medical Sciences, Chandorkar
3. Practical Clinical Biochemistry, Harold Varley
4. Medical Laboratory Sciences, Theory & Practical, A. Kolhatkar
5. Medical Laboratory Technology – Volume I, Kanai Mukherjee
6. Medical Laboratory Technology – Volume II, Kanai Mukherjee
7. Medical Laboratory Technology Methods & Interpretation, Ramnik Sood
8. Enzymes: Biochemistry, Biotechnology & Clinical chemistry, Palmer Trevor, Publisher: Horwood Pub. Co.
9. Outlines of Biochemistry, Erice Conn & Paul Stumpf; John Wiley and Sons
10. Fundamentals of Biochemistry, Donald Voet & Judith Voet, John Wiley and Sons, Inc
11. Lehninger, Principles of Biochemistry, David Nelson & Michael Cox, W.H. Freeman and company,
12. Biochemistry, Jeremy Berg, Lubert Stryer, W.H.Freeman and company
13. Biochemical Methods for Agricultural Sciences – Sadasivam and Manikam. Wiley Eastern Limited
14. Practical Clinical Biochemistry Harold Varley, CBS
15. An Introduction to Practical Biochemistry, David T Plummer. Tata McGraw-Hill Publishing Company Limited

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